

Final Exam Review session

Ways to solve ODEs:

- 1st order
- integrating factor (linear, first order) (2.1)
 - separable ($\frac{dy}{dx} = \frac{f(x)}{g(y)}$) (2.2)
 - exact equations & integrating factors (2.6)
 $M(x,y) + N(x,y)y' = 0$
 $\frac{d\psi}{dx} = M, \frac{d\psi}{dy} = N$, then $\psi(x,y) = c$ is soln.
exact if $M_y = N_x$

2nd order

- Characteristic equation method (3.1)
 - real roots
 - complex roots
 - repeated roots
- Variation of Parameters (3.6)
- Reduction of order (3.4)

→ Series solutions (5.2)

Systems of Eqs (7.4-7.6)

Matrix Exponential

Existence & Uniqueness

Gronwall Inequality

Continuous time dependence

Comparison Theorem

After the midterm

Extension of solutions & Finite-time blowup

Stability of fixed points (linear system)
(nonlinear system)

Lyapunov Function

Poincaré-Bendixson Theorem

Big Ideas of Material from After the Midterm

(the pre-midterm material can be seen in the midterm review notes)

Extending solutions & blow-up

$\dot{x} = f(t, x)$, f cts, locally Lipschitz in x

Gluing lemma - If two solutions ϕ_1, ϕ_2 are C_1 , and agree at a point t_0 (in $I_1 \cap I_2$)
then $\phi_1(t) = \phi_2(t)$ for $t \in I_1 \cap I_2$
and $\phi(t) = \begin{cases} \phi_1(t) & t \in I_1 \\ \phi_2(t) & t \in I_2 \end{cases}$

can "glue solutions together"

comes from E-U

Extension $\psi(t)$ solution to $\dot{x} = f(t, x)$ defined on (t_-, t_+) .
Can extend soln to $(t_-, t_+ + \epsilon)$ iff

(A) $\lim_{t \rightarrow t_+^-} |\psi(t)| < \infty$
(left limit)

(B) There exists a sequence of $t_n \rightarrow t_+^-$ st.
 $\phi(t_n) \rightarrow \psi(t_+) = \psi_0$, with $|\psi_0| < \infty$

Global solution? - Exists when $|f(t, x)|$ is bounded on all of $\mathbb{R}^2$

Finite-time blowup - when $|f(t, x)| \rightarrow \infty$ for finite t

Linear Stability

Autonomous system: $\vec{x}' = \vec{F}(\vec{x})$ (no t !)

Equilibrium: Points where $\vec{F}(\vec{x}) = 0$

2x2 system: $\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = \begin{pmatrix} f_1(x_1, x_2) \\ f_2(x_1, x_2) \end{pmatrix}$

x_1, x_2 plane called phase plane

A solution $x(t)$ gives trajectory $(t, x(t))$
The family of trajectories - phase portrait

Linear System: $\dot{x} = Ax$

A's eigenvalues: λ_1, λ_2 (real)

- a) $\lambda_1, \lambda_2 < 0$ stable
- b) $\lambda_1, \lambda_2 > 0$ unstable
- c) either λ_1 , or $\lambda_2 > 0$ saddle

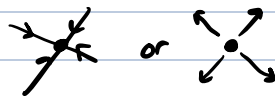
$\lambda_1, \lambda_2 = \alpha \pm i\beta$ (complex)

- a) $\alpha < 0$ spiral sink stable
- b) $\alpha > 0$ spiral source unstable
- c) $\alpha = 0$ center periodic orbit

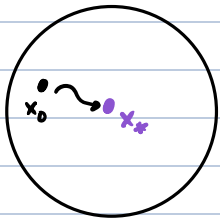
Repeated eigenvalues

- a) Two eigenvectors
- b) One eigenvector

lines straight through origin
improper / degenerate node

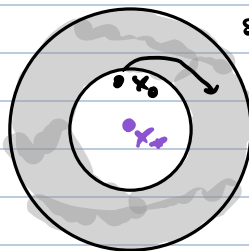


Asymptotically stable



Stable / Lyapunov stable

starts in neighborhood of size δ ,
stays in " of size ϵ for all time.



Unstable - not stable

Linear Stability Theorem

$$\dot{x} = Ax \quad x \in \mathbb{R}^n$$

IF all $\text{Re}(\lambda_i) < 0$ - asymptotically stable

IF any $\text{Re}(\lambda_i) > 0$ - unstable

IF all $\text{Re}(\lambda_i) \leq 0$ - Lyapunov stable

Decay estimate

$$\|e^{At} x_0\|_\infty \leq C e^{-\alpha t} \|x_0\|_\infty$$

Non-linear stability thm

IF all eigenvalues of A satisfy $\text{Re}(\lambda_i) < 0$, then the equilibrium x^* is asymptotically stable for the non linear system

(Proof in Lec 14 notes)

Lyapunov stability

L function, differentiable

$$L(x^*) = 0, \quad L(x) > 0 \text{ for } x \neq x^*$$

$$\dot{L}(x) \leq 0 \text{ for all } x \quad \text{Lyapunov stable}$$

$$\dot{L}(x) < 0 \text{ for all } x \neq x^* \quad \text{Asymptotically stable}$$

(Proof idea in notes)

How to build Lyapunov functions:

$$\rightarrow \text{Try Quadratic forms: } L(x, y) = ax^2 + bxy + cy^2 \quad a > 0, c > 0, 4ac > b^2$$

Alpha & Omega limit sets

$$\dot{x} = F(x) \quad \text{with initial cond } x(0) = x_0$$

ω -limit set of x_0

$$\omega(x_0) = \{y \in \mathbb{R}^2 : \exists t_n \rightarrow +\infty \text{ with } x(t_n; x_0) \rightarrow y\}$$

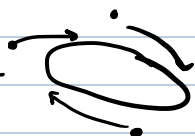
all points trajectory converges to in forward time

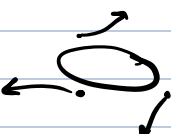
α -limit set of x_0

$$\alpha(x_0) = \{y \in \mathbb{R}^2 : \exists t_n \rightarrow -\infty \text{ with } x(t_n; x_0) \rightarrow y\}$$

all points trajectory converges to in backward time

Limit cycle - isolated closed trajectory (with a periodic orbit)

stable limit cycle: 

unstable limit cycle: 

(note, will be stable by going in backwards time)

For x_0 on a periodic orbit, $\omega(x_0) = \alpha(x_0) = \Gamma$ (the orbit)

Poincaré - Bendixson Thm

Let $x(t)$ be the trajectory whose orbit stays in a closed, bdd region.
Suppose the region contains finitely many equilibrium pts.

Then, ω -limit set is

- 1) an equilibrium pt
- 2) a periodic orbit
- 3) combinations of these (homoclinic/heteroclinic orbits)

